CPE 381 Classwork 14: Laplace Transform

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: October 28, 2026 (In-Class)

1. Write down the Laplace transform's integral formula. (9 Point)

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

2. Using the above integral formula for Laplace transform, computer the Laplace transform of a ramp signal $r(t) = tu(t)$.

   Use integration by parts:

$$\int udv = uv - \int vdu$$

   Use $u = t, dv = e^{-st}dt$ $\int dv = v = -\frac{1}{s}e^{-st}$

(1 Point)

$\int udv = \int t e^{-st} dt$

$R(s) = \int_{-\infty}^{\infty} t u(t) e^{-st} dt = \int_{0}^{\infty} t e^{-st} dt$

$= -\frac{t}{s} e^{-st}\Big|_0^{\infty} - \int_0^{\infty} -\frac{1}{s} e^{-st} dt$

$= -\frac{t}{s} e^{-st}\Big|_0^{\infty} + \frac{1}{s}\int_0^{\infty} e^{-st} dt$

$= -\frac{t}{s} e^{-st}\Big|_0^{\infty} + \frac{1}{s} \cdot \frac{1}{-s} e^{-st}\Big|_0^{\infty}$

$= -\frac{t}{s} e^{-st}\Big|_0^{\infty} - \frac{1}{s^2}[0-1]$

$= -\frac{t}{s}e^{-st}\Big|_0^\infty + \frac{1}{s^2}$

$= \lim_{t\to\infty} -\frac{t}{s}e^{-st} - \left(-\frac{t}{s}e^{-st}\right)\Big|_{t=0} + \frac{1}{s^2}$ $\quad ((uv)' = u'v + uv')$

$= \lim_{t\to\infty} -\frac{t}{s}e^{-st} + 0 + \frac{1}{s^2}$

Apply L'hopital Rule →

$\lim_{t\to\infty} \frac{d}{dt}\left(\frac{t}{s}e^{-st}\right)$

$\lim_{t\to\infty} \left[1\cdot e^{-st} + t(-se^{-st})\right]$

$= \lim_{t\to\infty} \left[e^{-st}(1-st)\right]$

$= 0.$

$= 0 + 0 + \frac{1}{s^2} = \frac{1}{s^2}$

Hence $R(s) = \mathcal{L}[r(t)] = \frac{1}{s^2}$

3. Give that $x(t) = e^{at}u(t)$ has Laplace transform

$$X(s) = \frac{1}{s-a}$$

. Find the Laplace transform of $y(t) = \sin(at)u(t)$.

$\sin(at) = \frac{e^{jat} - e^{-jat}}{2j}$

So $\sin(at)u(t) = \frac{1}{2j}\left(e^{jat}u(t) - e^{-jat}u(t)\right)$

$\mathcal{L}[\sin(at)u(t)] = \frac{1}{2j}\left[\frac{1}{s-ja} - \frac{1}{s+ja}\right]$

$= \frac{1}{2j}\left[\frac{s+ja - s+ja}{s^2 - (ja)^2}\right] = \frac{1}{2j}\left[\frac{2ja}{s^2+a^2}\right]$

$= \frac{a}{s^2+a^2}$